

NEURAL ORCHESTRATOR

Integration Manual



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Version control

VERSION CONTROL			
Document history	Version	Date	Change Description
	1.0	29/09/2021	First version
	2.0	07/04/2020	Timestamp in XMLEvent (JL)
	2.1	07/04/2020	Class ids
	2.2	13/02/2023	Extended XML. DB integration.
	3.0	17/01/2025	Heatbeat. Event integration
	3.1	30/11/2025	Event integration update

Introduction

This document is intended to assist the user on how to integrate with the Neural Orchestr[ai]tor for:

- Retrieve heartbeat status XML
- Event integration

Heartbeat

An XML message is sent periodically to an IP:Port with a configurable time to notify its status to an external program.

```
<?xml version=1.0 encoding=utf-8?>
<HeartBeat>
  <DateHour>TimeStamp</DateHour>
  <FreeSpace>FreeSpace</FreeSpace>
  <TotalSpace>TotalSpace</TotalSpace>
  <ComputerId>0</ComputerId>
  <InferenceEngine>
    <Models>
      <Model>
        <Id> CameraID </Id>
        <Name> ModelType </Name>
        <Workers> WorkerID </Workers>
        <InferenceAvgTime>XX</InferenceAvgTime>
      </Model>
    </Models>
  </InferenceEngine>
  <Cameras>
    <Camera>
      <Id> CameraID </Id>
      <Name> Camera name </Name>
      <Status> X </Status>
      <FPS> X </FPS>
      <Resolution> XxY </Resolution>
    </Camera>
  </Cameras>
  <Alarms>
    <Alarm>
      <Id> AlarmID </Id>
      <Name> Alarm name </Name>
      <DateTime> TimeStamp </DateTime>
      <CameraId> CameraID</CameraId>
      <CameraName> Camera name </CameraName>
    </Alarm>
  </Alarms>
</HeartBeat>
```

Where:

DateHour: XML date and time.

FreeSpace: Free space of the device.

TotalSpace: Total space of the device.

ComputerId: Device number identification.

InferenceEngine: Engine information for each camera.

- Id: camera identification.
- Name: Model number identification
- Workers: model worker identification.
- InferenceAvgTime: model average inference time.

Camera: devices information.

- Id: camera number identification.
- Name: Vamera name.
- Status: number that identify camera connection status
 - CONNECTED = 1
 - NOT_CONNECTED = 2
 - CONNECTING = 3
 - CONNECTION_LOST = 4
 - RECONNECTING = 5
- FPS: camera input frame
- Resolution: camera resolution in pixels

Alarms: Alarm information for camera.

- Id: Alarm ID
- Name: Alarm name
 - IMAGE_TOO_BLURRY
 - IMAGE_TOO_DARK
 - MOTION_ALARM
 - IMAGE_TOO_BRIGHT
 - GLOBAL_SCENE_CHANGE
 - SIGNAL_LOSS
 - MOTION_ALARM
- DateTime: Alarm status date and time.
- Camerald: camera number identification that trigger the alarm.

In the settings section of the Neural Orchestr[ai]tor interface you can activate and configure the heartbeat:

Heartbeat

Enabled ☒ Heartbeat rate [s]:

IP: Port:

Event integration

Events received from Neural Orchestr[ai]tor are summarised in a DB table, and these can be periodically sent in a JSON message over HTTP to a user-defined endpoint. A JSON message will be sent with the total events for each classification and camera in that time interval.

The interval is configurable, and can have these values: 60, 30, 15, and 5 minutes.

The default value is 60 minutes.

JSON message:

```
[
  {
    "name": "Camera 1",
    "DateTime": "2024-09-24T10:15:00",
    "IdComputer": 1,
    "IdLocalization": 3,
    "IdCamera": 1,
    "EventName": "Event Name 1",
    "IdClassification": 2,
    "TotalCount": 230,
    "SpeedCount": 230,
    "SumSpeed": 0,
    "AvgSpeed": 26
  },
  {
    "name": "Camera 1",
    "DateTime": "2024-09-24T10:15:00",
    "IdComputer": 1,
    "IdLocalization": 3,
    "IdCamera": 1,
    "EventName": "Event Name 2",
    "IdClassification": 2,
    "TotalCount": 242,
    "SpeedCount": 242,
    "SumSpeed": 0,
    "AvgSpeed": 25
  },
  {
    "name": "Camera 2",
    "DateTime": "2024-09-24T10:15:00",
    "IdComputer": 1,
    "IdLocalization": 3,
    "IdCamera": 2,
    "EventName": "Event Name 3",
    "IdClassification": 2,
    "TotalCount": 236,
    "SpeedCount": 236,
    "SumSpeed": 0,
    "AvgSpeed": 14
  },
  {
    "name": "Camera 2",
    "DateTime": "2024-09-24T10:15:00",
    "IdComputer": 1,
```

```

    "IdLocalization": 3,
    "IdCamera": 2,
    "EventName": "Event Name 4",
    "IdClassification": 2,
    "TotalCount": 236,
    "SpeedCount": 236,
    "SumSpeed": 0,
    "AvgSpeed": 23
  }
]

```

Object	ClassID
MOTORCYCLE	1
CAR	2
VAN	3
TRUCK	4
BUS	5
PERSON	6
BICYCLE	7
ABANDONED_OBJECT	8
ELECTRIC_SCOOTER	9
SMOKE	10
TRICYCLE	14
ANIMALS	114 - 123
FACE	201
FOG	202
GUN	301
KNIFE	302